BR

Bulk Repair

INVENTORY PORTAL

Complete User Guide

Har tab, har screen, har field — step-by-step Hinglish guide. Inventory, imports, rules, reports, purchasing, security aur troubleshooting.

Edition	Version	Date	Applies to
Hinglish	1.0	19 October 2026	Portal · Desktop · Android

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Bulk Repair — Complete User Guide (Hinglish)

Ye guide app ke **har tab aur har screen** ko detail me explain karta hai: kaun use kar sakta hai, kya fields hain, step-by-step kya karna hai, aur kahan galti hoti hai.

> **Legend:** ✅ = Local mode aur Apps Script mode dono me kaam karta hai. ⚠️ = sirf Apps Script / Google Sheets mode me available hai.

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1. Is guide ko kaise use karein

- Pehli baar app chala rahe ho? Section 3 → 4 → 5 padho.
- Roz ka kaam karna hai? Section 7 (Inventory), 8 (Imports), 10 (Reports) kaafi hai.
- Admin ho? Section 17, 18, 19 zaroor padho.
- Kuch kaam nahi kar raha? Section 23 (Troubleshooting) dekho.

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2. App kya karta hai

Bulk Repair ek repair/warehouse inventory system hai:

- Devices ka master data (barcode, model, brand, status, age, location, priority)
- Excel/CSV/PDF se bulk import aur rollback
- Multiple sources ka merge (Apps Script mode)
- Rules engine: age/status/location se auto bucket + priority
- Dashboard, ageing trend, reports (CSV/XLSX/PDF)
- Barcode labels printing
- Threshold alerts (email/WhatsApp)
- Vendors aur purchase orders, receiving se stock update
- Users, roles, sessions, audit log, 2FA

Do modes:

MODE	DATA KAHAN	KAB USE KAREIN
Local (offline)	`server/data/bulkrepair.db` (SQLite)	Single office/warehouse, fast, no internet
Apps Script	Google Sheets	Phone se kahin se access, Google ecosystem

Kuch advanced tabs sirf Apps Script mode me kaam karte hain — unpe ⚠️ mark hai. Local mode me Sources, Mapping, Priority rules, Report templates, Conflicts, Alerts checks, Reports definitions, Source data, Selftest aur file-based Multi-source merge sab supported hain. Sirf Google Sheets-specific cheezein (Sheet se direct import, Sheet security, Telegram test) Apps Script mode me rehti hain.

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3. Install aur start

3.1 Zaroorat

- Node.js 22+
- npm 10+
- Browser (Chrome/Edge/Safari)
- Android build ke liye: Android SDK + Java 21

3.2 Server + portal (local mode)

```
cd inventory-automation/server
npm install
npm run dev
```

- `npm run dev` = TypeScript build + server start
- URL: **http://localhost:5050** (port `server/.env` se)
- Pehli baar `server/data/bulkrepair.db` khud banta hai
- Migrations automatic chalti hain

3.3 Production build

```
cd inventory-automation/server
npm install
npm run build:panel   # portal build + server/public me copy
npm run build         # server build
npm start
```

3.4 Docker

```
cd inventory-automation
docker compose up --build
```

- App: `http://localhost:5000`
- Data `/data` volume me

3.5 Desktop app (self-contained)

`dist-builds/` me:

- macOS: `BulkRepair-1.0.0-macos-arm64.zip`
- Windows x64: `BulkRepair-1.0.0-windows-x64-portable.exe`

Desktop app **apna server khud start karta hai** — alag se `npm start` ki zarurat nahi.

- App launch → embedded server port **5050** pe (busy ho to 5051, 5052…)
- Database:

  - macOS: `~/Library/Application Support/Bulk Repair/data/bulkrepair.db`
  - Windows: `%APPDATA%\Bulk Repair\data\bulkrepair.db`
- Backups: usi folder ke `data/backups/`
- Server logs: `logs/server.log` (same app data folder)

Agar external backend (VPS/LAN) use karna ho to Setup me **Server URL** do — wahi URL embedded server ke bajaye use hoga.

3.6 Android APK

```
adb install -r BulkRepair-1.0.0-android-debug.apk
```

- USB: `adb reverse tcp:5050 tcp:5050` → app me `http://localhost:5050`
- Wi-Fi: `http://<Mac-ka-LAN-IP>:5050`

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4. First-time setup (3 steps)

Step 1 — Backend choose karo

OPTION	KAB
This computer (offline)	Local SQLite, no internet
Google Apps Script / Google Sheets	Data sheet me, phone se access

Apps Script choose kiya to `/exec` URL daalo → **Test connection**.

Step 2 — Google Sheet link (optional)

Sirf quick-access ke liye save hota hai. Local mode me skip kar sakte ho.

Step 3 — Owner account

- Email
- Password: **10+ characters**, letters + numbers dono
- Confirm password
- **Finish setup**

Ye account `OWNER` banega — sab permissions.

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5. Login, roles aur navigation

5.1 Login

- Email + password
- 2FA on hai to 6-digit code

- 5 galat attempts → 15 minute lock
- "Keep me signed in" = 30 din wala session

5.2 Roles

ROLE	ACCESS
VIEWER	Read-only: dashboard, inventory, reports dekh sakta hai
OPERATOR	Inventory add/edit, imports, labels, scan, AI commands
MANAGER	+ reports, scheduled reports, purchasing, conflicts clear
ADMIN	+ users, settings, sessions, vendors, 2FA reset
OWNER	Sab kuch

Sidebar options role ke hisaab se hide hote hain.

5.3 Sidebar groups

GROUP	TABS
Overview	Dashboard
Operations	Inventory, Imports, Labels, Reports, Scheduled reports, Alerts, Purchasing
Configuration	Sources, Field Mapping, Rules
Administration	Users & Security, Sessions & Audit, Activity & Audit, Settings
Tools	AI Assistant

Sidebar ke bottom me **language** selector (English / हिन्दी / ગુજરાતી) aur **Collapse** button hai.

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6. Dashboard

Kaun: sab roles.

6.1 Kya dikhta hai

BLOCK	MATLAB
Inventory items	Total rows
Active sources	Kitne sources on hain
Active reports	Report definitions
Recent errors	Latest import errors ka count
By status	Status distribution (bar)
By age bucket	Age buckets (donut)
By status bucket	Bucket distribution (bar)
Top brands	Top 5 brands (bar)
Ageing trend	Daily snapshots (line) — daily process chalne pe banta hai
Recent runs	Latest import/process jobs
Recent errors	Last errors with barcode/field/message

6.2 Actions

- **Refresh** — data reload
- **Run daily process** (OPERATOR+) — rules dobara apply, inventory rebuild

6.3 Kab kya karein

- Subah: errors aur ageing trend dekho
- Charts empty hain? Inventory import karo ya daily process chalao
- Recent errors me same barcode baar-baar? Us row ka data check karo

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7. Inventory

Kaun: sab dekh sakte hain; OPERATOR+ edit.

7.1 Toolbar (left → right)

CONTROL	KAAM
Search	Barcode / model / brand me search (debounced)
Bulk edit	Selected rows ka status/location/bin/priority badlo
Scan	Camera se barcode scan → search me bhar jaata hai
Status / Age / Brand / Category / Priority / Bin / Source	Filter dropdowns
Clear	Search + filters reset
Views	Saved filter presets
Columns	Columns hide/show
Export	CSV export (columns choose karke)
Add row	Naya item

7.2 Saved views (Views menu)

1. Search/filters set karo
2. **Views** → **Save current view…**
3. Naam do (e.g. "Apple — above 75 days") → **Save view**
4. Dobara use karna ho to **Views** → **naam** pe click
5. Delete: usi row me **X**

Search, filters aur page size automatically save rehte hain — reload pe wapas milte hain.

7.3 Columns menu

- Checkbox se column on/off
- **BARCODE** column lock hai (hide nahi hota)
- **Show all columns** se sab wapas
- Selection browser me save rehta hai

7.4 Row detail drawer

Row pe click karo:

- Saare fields
- **History** timeline:
  - `Added to inventory`
  - `Removed from inventory`
  - `FIELD: old → new` (status, location, bin, priority, age, model, brand)
- Buttons: **Edit**, **Delete**, **Close**

History har write path se banti hai — REST, imports, bulk edit, purchases.

7.5 Naya item

Add row → form:

FIELD	NOTE
BARCODE	Required, unique
MODEL / BRAND	Text
STATUS	Raw status
STOCK_AGE	Days
AGE_BUCKET / PRIORITY / STATUS_BUCKET	Rules se auto (manual override bhi ho sakta hai)
LOCATION / BIN_BUCKET	Rack/bin
SOURCE_TYPE	Manual entries ke liye

Save pe rules apply hote hain.

7.6 Edit / Delete

- **Edit:** drawer → Edit → fields badlo → Save
- Barcode editable nahi hota
- **Delete:** drawer → Delete → confirm. Ye permanent hai (history me `Removed` record rehta hai).

7.7 Bulk edit

1. Rows select karo (checkbox)
2. **Bulk edit**
3. Fields choose karo: status, status bucket, location, bin bucket, priority
4. Apply

Sirf selected fields badalte hain; baaki data same rehta hai. Action audit log me jaata hai.

7.8 Export CSV

1. **Export**
2. Columns tick karo (All / None shortcuts)
3. **Export CSV** → download

7.9 Common mistakes

- Barcode duplicate → save fail (barcode unique hai)
- Empty filter result → **Clear** dabao
- Purana data dikh raha hai → **Refresh**

7.10 Hardware barcode scanners (desktop)

Desktop (Windows/macOS) pe camera scan nahi, **USB hardware scanner** support hai.

- Zyada tar USB scanners **keyboard ki tarah** kaam karte hain — koi driver install nahi karna padta.

- Inventory page pe kahin bhi scan karo — search box khud bhar jaata hai.
- **Settings** → **Barcode scanner** me:
  - Mode: Hardware / Camera (desktop pe hardware)
  - Minimum code length (chhoti typing ignore hoti hai)
  - Terminator: Enter / Tab / None
  - **Test scanner** field me scan karke verify karo
- **Serial/COM scanners:** Settings → **Connect serial scanner** (desktop app me Web Serial support hai, 9600 baud default).
- Mobile app me camera scan available rehta hai.

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8. Imports

Kaun: OPERATOR+ (commit/rollback); VIEWER dekh sakta hai.

Imports ke **5 tabs** hain.

8.1 Universal import ✅

File se data laane ka main rasta.

Supported files: CSV, TSV, TXT, XLS, XLSX, PDF (max 15 MB / 50,000 rows).

Step-by-step:

1. **Imports** → **Universal import**
2. **Choose file** — file select karo
3. App columns auto-detect karta hai
4. Mapping check karo:
   - Har source column ke saamne target field
   - `BARCODE` required hai
   - Galat mapping ho to dropdown se badlo
5. **Validate**:
   - Green = ready
   - Red rows = errors (row number + message)
6. **Import now**
   - Preview me total/valid/imported/failed counts
   - Partial failure ho sakta hai — failed rows list dekho
7. Zarurat ho to **Roll back import**

Result card:

FIELD	MATLAB
Status	Committed / Dry run
Processed	Total rows
Valid	Rules pass
Imported	Inventory me gaye
Failed	Reject hue
Skipped	Empty/duplicate
Errors table	Row + message

Rollback kya karta hai:

- Is job ne jo naye barcodes banaye → delete
- Jo pehle se the → unki purani values restore
- Dobara rollback nahi hota
- Sirf committed job ka rollback hota hai, dry run ka nahi

Tips:

- Pehle 10-20 rows ki test file se mapping verify karo
- Barcode column me spaces na ho
- Age column numeric ho
- Ek hi file me duplicate barcode na ho

8.2 Multi-source ✅

Purpose: QC Stock + TRC + Inventory + Location jaise alag sources ko **barcode pe join** karke ek final record banana.

Local mode (files):

1. Har source ke liye ek CSV/Excel file choose karo (QC Stock, TRC, Inventory, Location)
2. **Validate files** — har file stage hoti hai, columns auto-map hote hain
3. **Import all sources** — merge chalta hai
4. Result: devices, inserted, updated, conflicts
5. Galti ho to **Roll back import**

Apps Script mode (sheets):

1. Har source ka type + sheet name + sheet ID do
2. **Validate all** → **Import all sources**

Conflict ka matlab: same barcode, same field, do sources me alag value. Kaun jeetega ye **Priority rules** decide karte hain (local me bhi editable).

8.3 Conflicts ✅

Jab multi-source merge me do sources ki value alag hoti hai, wo yahan aati hai.

COLUMN	MATLAB
Barcode	Device
Field	Kaunsa field
Source A / Value A	Jeetne wala
Source B / Value B	Doosra
Final value	Jo inventory me gaya
Rule	Kaunsa rule laga

Actions:

- **Refresh** — latest conflicts
- **Clear** — saare conflict rows hatao (inventory data nahi badalta)

Conflict dekhkar priority rules adjust karo, phir merge dobara chalao.

8.4 Priority rules ✅

Batata hai ki merge ke waqt kaunsa source jeetega. Local mode me bhi ab editable hai.

FIELD	MATLAB
Field	Kaunsa data field (status, location, etc.)
Source type	QC_STOCK / TRC / INVENTORY / LOCATION
Rank	Chhota number = zyada priority

Example:

FIELD	SOURCE	RANK
status	QC_STOCK	1
status	TRC	2
status	INVENTORY	3

Matlab: status ke liye QC_STOCK sabse bharosemand.

Local mode me ye rules save hote hain (merge workflow khud Apps Script mode me chalta hai).

8.5 Source data ✅

Consolidated source rows ka raw view.

- **Consolidated source rows** — search + table (max 500 rows shown)
- **Import history** — last 20 runs (source, rows, status)

Ye tab debug ke liye hai: import ke baad data kaisa aaya, ye verify karo.

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9. Labels

Kaun: OPERATOR+.

9.1 Settings

SETTING	OPTIONS
How many labels	1–200
Label size	50×25, 38×25, 60×30 mm
Columns	2, 3, 4
Barcode height	20–80
Fields	Barcode text / Brand + model / Location + bin

Settings browser me save hoti hain.

9.2 Print

1. Settings set karo
2. **Print**
3. Print dialog:
   - Margins: **None**
   - Scale: **100%**
   - Background graphics: **ON**
4. Preview check karo → Print

9.3 Common mistakes

- Barcode cut ho raha hai → barcode height kam karo ya label size badlo
- 3 columns me fit nahi → 2 columns karo
- Blank labels → inventory me barcode missing hai

---

10. Reports

Kaun: MANAGER+ (VIEWER dekh sakta hai).

10.1 Report definitions

Table me saved report definitions.

Actions:

- **New report** — naam + group by + filters
- **Edit** (pencil)
- **Delete** (trash)

- **Generate now** — turant run

10.2 Generate now

FIELD	MATLAB
Group by	status / brand / age_bucket / status_bucket / location / bin_bucket / model
Filters	status, brand, age bucket, status bucket, location

Result: groups + counts.

10.3 Templates ✅

Reusable report formats (advanced; local mode me bhi supported):

- **New template** — naam + config JSON
- **Edit / Delete**
- **Config JSON** structure aapke template engine pe depend karta hai

10.4 Exports

BUTTON	OUTPUT
CSV	Grouped counts
Excel	XLSX with filters
PDF	Print-ready report

Export all to sheets ⚠️ — Apps Script mode me sab reports sheets me likhta hai.

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11. Scheduled reports

Kaun: MANAGER+ dekh sakta hai, ADMIN bana sakta hai.

11.1 New schedule

FIELD	MATLAB
Name	Internal naam
Group by	Report dimension
Frequency	Daily / Weekly / Monthly
Format	PDF / XLSX
Recipients	Comma-separated emails
Status / Brand / Location filter	Optional exact match
Active	ON = automatic send

11.2 Kaise chalta hai

- Server boot pe + har **30 minute** check karta hai
- Due schedule (last run + period) → report generate → email
- `last_run_at` save hota hai — restart pe duplicate nahi jaata
- Email fail ho to schedule active rehta hai, next cycle me retry

11.3 Actions

- **Run** — abhi bhejo
- **Edit** — settings badlo
- **Delete** — band karo

11.4 Common mistakes

- Email nahi ja raha → Settings me SMTP configure karo, **email test** dabao
- Recipients comma se separate hone chahiye
- "Never" last run = pehli baar abhi tak due nahi hua

---

12. Alerts

Kaun: ADMIN bana sakta hai; sab dekh sakte hain.

12.1 Alert types

TYPE	MATLAB	EXAMPLE
AGE_OVER	Kitne din se purana stock	90+ days ka stock
STATUS_BUCKET_COUNT	Bucket me count threshold	QC_FAIL > 50
MISSING_FIELD	Field missing	location khali

12.2 New alert

FIELD	NOTE
Name	Unique
Type	Upar wale 3
Threshold	Number
Filter field / value	Optional (brand = Apple)
Email to	Recipients
Active	ON/OFF

12.3 Testing

Card **Notification tests** me:

- **Send test email**
- **Send WhatsApp test**

Dono Settings ki config use karte hain. Fail ho to exact error message dikhta hai.

12.4 Check now

Check now — abhi rules evaluate karo, triggered alerts dikhenge. Daily process ke saath bhi automatic chalta hai.

---

13. Purchasing

Kaun: MANAGER+.

Do tabs: **Purchase orders** aur **Vendors**.

13.1 Vendors tab

FIELD	NOTE
Name	Required, unique
Contact person	Text
Email	Text
Phone	Text
Address	Text
Notes	Text
Active	ON/OFF

Actions: **Edit**, **Delete**. Vendor delete tabhi hoga jab uska koi PO na ho.

13.2 Purchase orders tab

Table columns: PO number, vendor, status, lines, ordered, received, expected.

Status flow:

```
DRAFT → ORDERED → PARTIAL → RECEIVED
              ↘ CANCELLED
```

13.3 Naya PO

1. **New purchase order**
2. Vendor choose karo
3. Status: **Draft** ya **Ordered**
4. Expected date + notes
5. **Add line** — har line me:
   - Barcode (optional)
   - Model
   - Brand
   - Quantity
   - Unit cost
6. **Save order**

PO number auto-generate hota hai (`PO-YYYYMMDD-001`).

13.4 Receive

1. PO row me **Receive**
2. Har line ka remaining quantity dikhta hai
3. **Receive now** me quantity daalo (0 = skip)
4. **Receive stock**

Result:

- Received quantity update
- Barcode wali line inventory me `IN_STOCK` ban jaati hai
- Status auto: sab aa gaya → `RECEIVED` , partial → `PARTIAL`
- Over-receive reject hota hai

13.5 Cancel

- Sirf un orders ke liye jisme kuch receive nahi hua
- `CANCELLED` ke baad edit nahi

13.6 Common mistakes

- "Cannot receive N; only M remain" → quantity remaining se zyada hai
- Received order edit nahi hota — naya PO banao
- Vendor delete nahi ho raha → pehle uske POs cancel/delete karo

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14. Sources

Kaun: ADMIN.

Source = data ka origin (QC sheet, TRC sheet, inventory sheet, location sheet).

14.1 New source

FIELD	NOTE
Name	Unique
Role	PRIMARY (main data) ya JOIN (extra data)
Type	Free text, e.g. Google Sheet; system QC/TRC/Inventory/Location se source type infer karta hai
Sheet name	Google Sheet tab ka naam
Sheet ID	External spreadsheet ho to
Join key	Default `barcode`
Join fields	Comma-separated fields is source se lene hain
Active	ON/OFF

14.2 Source types ka matlab

TYPE	USE
QC_STOCK	QC pass/fail data
TRC	Repair/TRC data
INVENTORY	Main inventory sheet
LOCATION	Location/bin mapping
OTHER	Manual/import source

14.3 Kab kya karein

- Naya sheet aaya → source add karo → mapping karo
- Source band karna hai → Edit → Active OFF
- Delete local mode me nahi (Apps Script me available)

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15. Field Mapping

Kaun: ADMIN.

Source columns ko standard fields se jodta hai.

15.1 Kaise map karein

1. Source choose karo
2. Har row me:
   - **Source column** (sheet ka header)
   - **Target field** (`BARCODE`, `MODEL`, `BRAND`, `STATUS`, `STOCK_AGE`, `LOCATION`, ...)
   - **Required** checkbox
   - **Output order**
   - **Active**
3. **Save mapping**

15.2 Rules

- `BARCODE` required hai
- Ek source column ek hi target pe
- Empty target = ignore
- Mapping save karne ke baad import karo

15.3 Common mistakes

- Barcode mapping missing → import fail
- Age column text me hai → numeric karo
- Status ka naam galat → rules match nahi karenge

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16. Rules

Kaun: ADMIN.

Rules ka result: age bucket, status bucket, bin bucket, priority. Changes **next process run** pe apply hote hain.

16.1 Age buckets tab

FIELD	MATLAB
Min age	Days (inclusive)
Max age	Days (inclusive)
Bucket name	e.g. `Below 30`, `30–60`, `Above 75`
Priority	`P1`, `P2`, `P3`

Example:

MIN	MAX	BUCKET	PRIORITY
0	29	Below 30	P3
30	59	30-60	P2
60	74	60-75	P2
75	9999	Above 75	P1

Rule overlap na ho — pehla match jeetta hai.

16.2 Status buckets tab

FIELD	MATLAB
Raw status	Sheet me jo status aata hai (`L4` , `QC Fail`)
Bucket	Standard bucket (`READY` , `QC_FAIL`)
Marking	Optional note/flag

16.3 Location rules tab

FIELD	MATLAB
Pattern	Location ka pattern (`SW*` , `A1`)
Bucket	Bin bucket (`Shelf A` , `Cage 1`)
Condition	Optional extra condition

16.4 Priority rules

⚠️ Apps Script mode (Imports → Priority rules tab).

Merge ke waqt field-level winner decide karta hai. Rank chhota = zyada priority.

16.5 Apply karna

Rules save karne ke baad:

1. **Dashboard** → **Run daily process** (ya Imports ke baad process)
2. Inventory me buckets update ho jaate hain

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17. Users & Security

Kaun: ADMIN.

Do tabs.

17.1 Users tab

COLUMN	MATLAB
Email	Login
Role	VIEWER → OWNER
Active	YES/NO
Password	Set hai ya nahi
Locked until	Lockout time
Last login	Last successful login

New user:

1. **New user**
2. Email, role, password (10+ chars, letters + numbers)
3. Save

Edit user:

- Role badlo
- Active OFF = login block + existing sessions revoke
- Apna role khud change nahi kar sakte

Reset password:

- User row → reset
- Naya password set karo
- Us user ke saare sessions revoke ho jaate hain

2FA reset: agar user ne authenticator kho diya, ADMIN `auth.totpReset` se 2FA off kar sakta hai (Settings ke through).

17.2 Sheet security tab ⚠️

Sirf Apps Script mode. Google Sheet ke tabs lock/unlock karta hai taaki accidental edit na ho.

- **Lock** — sheet protect
- **Unlock** — protection hatao

Local mode me ye tab data nahi dikhayega.

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18. Sessions & Audit

Kaun: ADMIN.

18.1 Active sessions

COLUMN	MATLAB
User	Email
Role	Current role
Signed in	Session start
Last seen	Last activity
Expires	Session expiry

Actions:

- **Sign out** — sirf ye session revoke
- **All** — us user ke saare devices se sign out

Revoke hone ke baad us token se koi API call nahi chalegi.

18.2 Login history

Last 100 events:

EVENT	MATLAB
Login OK	Successful login
Login Failed	Wrong password
Login Locked	5 failures ke baad lock
Login Blocked Locked	Locked account se try
Logout	Manual logout

IP column se suspicious logins pakdo.

18.3 Kab dekhna chahiye

- Roz shaam ko failed logins
- Kisi user ke multiple devices
- Employee exit ke baad uske sessions revoke karo

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19. Settings

Kaun: ADMIN (VIEWER ke liye read-only).

19.1 System settings

Groups:

GROUP	KEYS
Telegram	`TELEGRAM_BOT_TOKEN`, `TELEGRAM_CHAT_ID`
Auto backup	`AUTO_BACKUP`, `BACKUP_FOLDER_ID`
SMTP	`SMTP_HOST`, `SMTP_PORT`, `SMTP_USER`, `SMTP_PASS`, `SMTP_FROM`, `ALERT_EMAIL_TO`
WhatsApp	`WHATSAPP_TOKEN`, `WHATSAPP_PHONE_ID`, `ALERT_WHATSAPP_TO`
Google	`GOOGLE_CLIENT_ID`, `GOOGLE_AUTO_CREATE`
LLM	`LLM_API_URL`, `LLM_API_KEY`, `LLM_MODEL`

Save karo → test buttons se verify.

19.2 Notification tests

- **Send test email** — SMTP check
- **Send WhatsApp test** — Cloud API check

19.3 Connection

- API URL (desktop/mobile ke liye)
- Sheet URL (Apps Script)
- Save

19.4 Two-factor authentication

1. **Enable 2FA**
2. Secret key ya `otpauth://` URI authenticator app me daalo
   - Google Authenticator → "+" → Enter setup key
3. App ka 6-digit code daalo
4. **Verify and enable**
5. **8 recovery codes** copy karke safe rakho
6. Disable: password + current code

Recovery code ek hi baar chalta hai.

19.5 Backup

- Boot pe + har 6 ghante
- `server/data/backups/`
- Latest 7 rakhe jaate hain

---

20. AI Assistant

Kaun: OPERATOR+.

Natural language command likho:

- `List Apple devices older than 75 days`
- `How many devices are in Bin Storage?`
- `Show today's inventory summary`
- Hindi bhi: `75 din se purane Apple devices dikhao`

Kaise kaam karta hai:

1. Command parse hoti hai (filters: brand, location, status, age)
2. Inventory query chalti hai
3. Result JSON me dikhta hai

LLM mode (optional): Settings me `LLM_API_URL` + `LLM_API_KEY` + `LLM_MODEL` set karo. Model sirf filters suggest karta hai; SQL khud server banata hai. LLM fail ho to rules parser chal jaata hai.

---

21. Roz ka workflow

Subah

1. Dashboard — errors, ageing, counts
2. Sessions & Audit — failed logins (admin)

Naya stock

1. Imports → Universal import → file
2. Validate → Import now
3. Errors check karo
4. Labels print karo

Movement

1. Inventory → search barcode
2. Row → Edit → location/status
3. History me change confirm karo

Bulk

1. Rows select → Bulk edit
2. Status/location/priority apply

Orders

1. Purchasing → New PO
2. Receive hone pe **Receive**
3. Inventory auto update

Shaam

1. Reports → PDF/XLSX
2. Scheduled report agar weekly chahiye

---

22. Backup aur restore

Backup

- Automatic: boot + har 6 ghante
- Location: `server/data/backups/`
- Retention: 7

Manual backup

- Settings → Auto backup ON
- Ya server band karke `data/bulkrepair.db` copy karo

Restore

1. Server band karo
2. `data/bulkrepair.db` rename karo
3. Backup file ko `bulkrepair.db` naam se copy karo
4. Server start karo

Import rollback

- Committed import ka **Roll back import**
- Dry run ka rollback nahi

---

23. Troubleshooting

PROBLEM	SOLUTION
Page nahi khul raha	`http://localhost:5050/api/health` check karo
API error	Setup → Connection me URL; desktop pe `http://localhost:5050`
Android connect nahi	Wi-Fi same? `adb reverse tcp:5050 tcp:5050`
Login fail	Caps Lock, password 10+ chars, admin reset
Account locked	15 min wait; admin lock clear kar sakta hai
2FA code galat	Phone time auto karo; recovery code use karo
2FA reset chahiye	Admin se `TOTP reset` karwao
Email nahi ja raha	SMTP host/from/pass; test email
WhatsApp nahi ja raha	Token, phone ID, country code
Import fail	Barcode column, 50k limit, errors preview
Rollback nahi dikh raha	Sirf committed job; dry run nahi
Charts khali	Inventory empty / daily process nahi chala
Camera nahi chal raha	Permission; HTTPS/localhost
Print bigad raha	Margins None, scale 100%, background ON
Sessions tab khali	Koi active session nahi ya admin nahi ho
Multi-source error	Har source ke liye file select karo; barcode column hona chahiye
Priority rules error	Field khali na ho; source type QC_STOCK/TRC/INVENTORY/LOCATION/OTHER me se ho
Sheet security error	Apps Script only
Report templates error	Ab local me supported hai — Config JSON valid hona chahiye

24. Security checklist

- [ ] Owner password strong (10+ chars, letters + numbers)
- [ ] 2FA enable karo
- [ ] Extra users deactivate karo
- [ ] Har hafte Sessions & Audit check karo
- [ ] Backups safe location pe
- [ ] Public internet pe expose kar rahe ho to HTTPS reverse proxy
- [ ] SMTP password/LLM key rotate karte raho

- [ ] Employee exit pe uske sessions revoke karo

25. Useful commands

```
# server start
cd server && npm run dev

# build only
cd server && npm run build

# tests
cd server && npm test

# panel rebuild
cd server && npm run build:panel

# admin user (CLI)
cd server && npm run create-user -- owner@example.com 'StrongPass123' ADMIN

# Docker
docker compose up --build

# Android APK
cd mobile/android && ./gradlew assembleDebug
```

26. FAQ aur glossary

FAQ

Data kahan save hota hai? Local: `server/data/bulkrepair.db`. Apps Script: Google Sheet.

Internet chahiye? Local mode me nahi. Apps Script me haan.

Multi-warehouse? Abhi single global inventory. Multi-warehouse implement nahi hua.

Hindi/Gujarati UI? Sidebar bottom me selector. Default English.

Data delete ho gaya? Backup restore ya import rollback.

Ek user kitne devices se login kar sakta hai? Koi limit nahi; Sessions page se revoke kar sakte ho.

Glossary

TERM	MATLAB
Barcode	Device ka unique ID
Bucket	Standard category (age/status/location)
Priority	P1/P2/P3 — urgency
Source	Data ka origin (sheet/file)
Mapping	Source column → system field
Staging	Import ka temporary area (commit se pehle)
Job	Ek import run ka record
Conflict	Do sources me same field ki alag value
Rollback	Committed import undo
Bin bucket	Physical bin/shelf group
Session	Login token
2FA	Password ke upar authenticator code